

THEORY

1. What is the difference between XLOOKUP, VLOOKUP, INDEX-MATCH, and when would you choose each in a production report?

2. What are the limitations of VLOOKUP when working with large business datasets?

3. What is the difference between exact match and approximate match, and where can an incorrect match create a business problem?

4. What is the difference between relative, absolute, and mixed cell references, and when would each be useful in a reporting model?

5. What is the difference between Excel Tables and ordinary cell ranges?

6. Why are structured references useful when building recurring reports?

7. What is the difference between a calculated column and an aggregated calculation in an Excel data model?

8. What is Power Query, and how is it different from traditional Excel formulas?

9. What types of data-cleaning operations are better handled in Power Query than with formulas?

10. What is query refresh in Power Query, and why is it important for recurring reporting?

THEORY

11. What is the difference between merging and appending queries in Power Query?

12. What is the purpose of PivotTables in data analysis?

13. What are the advantages and limitations of PivotTables compared with formula-based reports?

14. What is the difference between a PivotTable and a PivotChart?

15. What are slicers and timelines, and when would you use them in a management dashboard?

16. What is conditional formatting, and how can it be used for business monitoring rather than simple highlighting?

17. What is the difference between SUMIF and SUMIFS?

18. What is the difference between COUNTIF, COUNTIFS, COUNTA, and COUNT?

19. What is the difference between IF, IFS, AND, OR, and SWITCH, and when would you use them in business logic?

20. What are dynamic-array functions, and how do FILTER, SORT, UNIQUE, and SORTBY help analysts?

THEORY

21. What is the difference between a formula-driven dashboard and a PivotTable-driven dashboard?

22. What is Power Pivot, and when would you consider using it instead of standard Excel worksheets?

23. What is a data model in Excel?

24. What are relationships between tables, and why are they important when analyzing multiple datasets?

25. What is the purpose of a calendar or date table in analytical reporting?

26. What is the difference between a measure and a calculated column in the Excel Data Model?

27. What is data validation, and how can it prevent errors in operational Excel files?

28. What are named ranges, and when can they improve maintainability of a workbook?

29. What is the purpose of Excel's error-handling functions such as IFERROR and IFNA?

30. What is the difference between a blank cell, zero, empty text, and #N/A from an analytical perspective?

THEORY

- 31.** What are the common causes of duplicate records in Excel datasets?
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- 32.** What are the main types of data-quality problems an analyst should check before reporting?
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- 33.** What is reconciliation, and why is it important in management reporting?
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- 34.** What is variance analysis, and how is variance percentage different from absolute variance?
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- 35.** What is the difference between average, weighted average, median, and percentile?
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- 36.** Why can a simple average produce a misleading business conclusion?
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- 37.** What is a rolling average, and why would an analyst use one?
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- 38.** What is a cumulative total, and how is it different from a monthly total?
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- 39.** What is Pareto analysis, and how can it be implemented in Excel?
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- 40.** What is sensitivity analysis, and how can Excel be used for it?
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THEORY

41. What is scenario analysis, and how does it differ from sensitivity analysis?

42. What is the purpose of Goal Seek, Data Tables, and Scenario Manager?

43. What are the common reasons an Excel workbook becomes slow?

44. How do volatile functions affect Excel performance?

45. What are the risks of using too many external links in an Excel reporting environment?

46. What is workbook protection, and what should and should not be protected in an analytical workbook?

47. What is the difference between data cleaning, data transformation, and data analysis?

48. What validation checks should be performed before publishing an Excel dashboard?

49. What makes an Excel dashboard effective for senior management?

50. What factors should determine whether an analyst uses Excel, Power Query, or a BI tool for a particular task?

PRACTICAL

51. You receive 12 monthly Excel files from different teams. The column names, date formats, product names, and region values are inconsistent. How would you build a process to consolidate and standardize them without manually cleaning them every month?

52. Your manager says August revenue is ₹4.8 crore, while your Excel report shows ₹5.3 crore. How would you systematically reconcile the two figures and identify the source of the difference?

53. You have 800,000 transaction records and the workbook is extremely slow. How would you redesign the Excel workflow rather than simply deleting data?

54. Two datasets contain the same customer information, but customer IDs are inconsistent because of spaces, leading zeros, and different formats. How would you match them reliably?

55. A salesperson appears as 'Rahul Sharma', 'rahul sharma', 'Rahul Sharma ' and 'R. Sharma' across different files. How would you standardize the records before calculating performance?

56. Your monthly report contains 25 KPIs. After refreshing the data, three KPIs suddenly change by more than 40%. How would you determine whether this is a genuine business change or a reporting error?

57. You have actual sales and targets for 500 salespeople. Management wants to know which regions are responsible for the overall target shortfall. How would you analyze it?

58. A product's revenue increased by 25%, but its units sold decreased by 10%. How would you use Excel to investigate what happened?

59. Revenue increased by 30%, but profit decreased by 15%. What Excel analysis would you perform to identify the drivers?

60. Management tells you that sales are down 12% and asks, 'Tell me why, not just where.' How would you structure your Excel analysis?

PRACTICAL

- 61.** You have customer-level monthly spending for two years. How would you identify customers whose spending has declined for three consecutive months?
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- 62.** How would you identify customers who purchased in January, February, and March but stopped purchasing from April onward?
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- 63.** How would you identify the top 10% of customers responsible for the largest percentage of revenue?
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- 64.** Management wants to compare the behavior of the top 10% of customers against the remaining 90%. What metrics would you calculate?
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- 65.** You have transaction-level data and need to calculate average order value by customer segment and month. How would you build the analysis in Excel?
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- 66.** A business wants to identify products that have high sales but low margins. How would you create this analysis in Excel?
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- 67.** You have product sales by month. How would you identify products with three consecutive months of declining sales?
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- 68.** A promotional campaign started in April. How would you compare sales before and after the campaign and determine whether the campaign appears successful?
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- 69.** Sales increased after a discount campaign, but management wants to know whether the increase was actually caused by the discount. How would you analyze the available data?
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- 70.** You receive an Excel file containing duplicate transactions, but some duplicates are legitimate repeat purchases. How would you distinguish true duplicates from valid transactions?

PRACTICAL

71. A source file contains 5% missing customer IDs. How would you investigate the reason and determine whether those records can still be used?

72. You discover that 8% of transactions have invalid product codes. How would you identify, quantify, and report the impact of these records?

73. A stakeholder asks for a customer retention report. How would you define the retention logic and build the analysis in Excel?

74. You need to identify customers who have not purchased during the last 90 days. How would you build a refreshable Excel report?

75. You have employee attendance data and need to identify employees with unusually high absenteeism. How would you approach the analysis?

76. HR asks you to analyze employee attrition by department, location, tenure, and manager. How would you structure the Excel report?

77. You have daily sales data for two years. Management wants to identify seasonality and unusual sales spikes. How would you analyze it?

78. A regional manager claims that their region is underperforming because of low customer traffic. How would you test this claim using Excel data?

79. You have sales targets for each month, but some targets were revised during the year. How would you ensure your variance report uses the correct target version?

80. Your report contains data from three different systems, and each system uses different customer identifiers. How would you create a reliable consolidated report?

PRACTICAL

81. You need to combine hundreds of CSV or Excel files received every day into one reporting dataset. How would you automate this using Excel and Power Query?

82. A Power Query refresh works correctly on your laptop but fails when another analyst opens the workbook. How would you troubleshoot it?

83. A Power Query transformation suddenly produces fewer rows than expected after a new monthly file arrives. How would you investigate the issue?

84. You have a master product table and a daily transaction table. Some products in the transaction file do not exist in the master table. How would you identify and report these exceptions?

85. You need to create a management dashboard containing revenue, orders, customers, average order value, target achievement, and growth. How would you design the dashboard?

86. Management wants the dashboard to be filterable by region, product category, salesperson, and month. How would you structure the workbook?

87. Your dashboard works correctly, but users frequently change formulas accidentally. How would you protect the workbook while keeping necessary inputs editable?

88. A stakeholder says your dashboard is too complicated. How would you redesign it without removing important information?

89. Your PivotTable shows a total that does not match the source data. What checks would you perform?

90. A lookup formula returns incorrect customer information for several records even though the formula appears correct. How would you debug it?

PRACTICAL

91. You are using XLOOKUP and suddenly many records return #N/A. How would you determine whether the problem is with the lookup formula or the source data?

92. Your manager wants a report that automatically highlights products below 80% target achievement and regions with negative month-over-month growth. How would you build it?

93. You need to create a report where users select a month and the entire report updates automatically. How would you design it?

94. You have actual sales, budget, previous-year sales, and forecast data. How would you build a management report showing all four with meaningful variance analysis?

95. A report has been used by management for six months, and you discover that one formula was incorrect. How would you determine which historical reports and business decisions were affected?

96. Your manager asks you to reduce a manual reporting process from 4 hours to 20 minutes. How would you identify opportunities for Excel automation?

97. You inherit an Excel workbook containing 30 sheets, thousands of formulas, hidden rows, external links, and undocumented macros. How would you audit it before making changes?

98. A stakeholder gives you a vague request: 'Give me a report showing why customers are leaving.' What questions would you ask before starting the Excel analysis?

99. Your analysis contradicts the business team's assumption about why sales declined. How would you validate your analysis and communicate the finding without creating conflict?

100. Walk through an end-to-end Excel project you handled: raw data, data-quality checks, cleaning, transformation, analysis, dashboard, validation, business insight, and recommendation.